

LAB 2

REFERENCE TABLE: LAB_GRADES

std_id	name	major	section	project_marks	cgpa	submit_date
s001	Abir	CS	1	18.5	3.91	2018-09-15
s002	Nafis	CSE	1	20	3.86	2018-08-15
s003	Tasneem	CS	1	18	3.57	2018-09-18
s004	Nahid	ECE	2	16.5	3.25	2018-08-20
s005	Arafat	CS	2	20	4.0	2018-09-13
s006	Tasneem	CSE	1	17.5	3.7	2018-08-15
s007	Muhtadi	ECE	1	19	3.67	2018-09-16
s008	Farhana	CSE	2	15	2.67	2018-08-16
s009	Naima	CSE	2	20	3.7	2018-08-14

FUNCTIONS REFERENCE

Aggregate Functions	Scalar Functions
MIN()	UPPER()
MAX()	LOWER()
AVG()	LENGTH()
SUM()	CONCAT()
COUNT()	NOW()



GROUP BY USE ২২

CS	
CSE	



SQL QUERY FORMATS

```
6(/(&7 BB )520 BB :+(5( BBBB
```

```
6(/(&7 BB )520 BB :+(5( BBBB *5283 %<BBBB
```

```
6(/(&7 BB )520 BB :+(5( BBBB *5283 %<BBBB 25'(5 %<BBBB
```

- SELECT __ FROM __ WHERE __ GROUP BY __ HAVING__

PART B

1. Write a SQL query to retrieve the minimum CGPA from the table.

^{↑ SELECT}
`SELECT MIN(cgpa) FROM lab;`



minimum
2.7

2. Write a SQL query to retrieve the minimum CGPA and display the column with an alias name 'minimum'.

`SELECT MIN(cgpa) AS minimum FROM lab;`

3. Write a SQL query to retrieve the last submission date from the table.

`SELECT MAX(sub_date) FROM lab;`

↪ 2018-10-01
Year Month Day

PART C: GROUP BY

1. Retrieve total number of students from each major.

```
SELECT COUNT(*) FROM lab GROUP BY major;
```

2. For each major find the minimum and maximum CGPA/Project marks, but only if there were (at least 2 students) in the major.
- * GROUP BY attribute and condition is HAVING

```
SELECT max(cgpa), min(cgpa) FROM lab GROUP BY major  
HAVING COUNT(*) >= 2;
```

3. For each major find the minimum and maximum CGPA/Project marks, but (consider only students who submitted before or on 15th sep.)
- where

```
SELECT max(cgpa), min(cgpa) FROM lab WHERE sub-date <= '2025-09-15'  
GROUP BY major;
```

Nested Queries

* Two queries needed

* Comparison

L1 →

Nested	CSE	44

Lab L1

L2 →

	CSE	45

Lab L2

ANY
↑

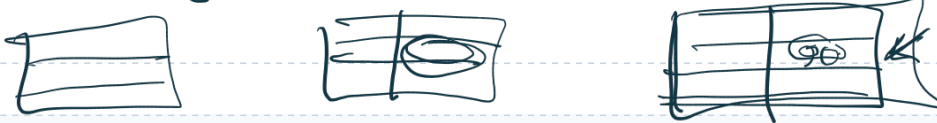
PART D: ATLEAST, ALL, EXISTS

1. Retrieve the CSE students whose CGPA/~~Project marks~~ is higher than at least 1 CS student (using ANY).

SELECT * FROM lab WHOSE major = 'CSE' WHOSE cgpa >
ANY(SELECT cgpa FROM lab WHERE major = 'CS');

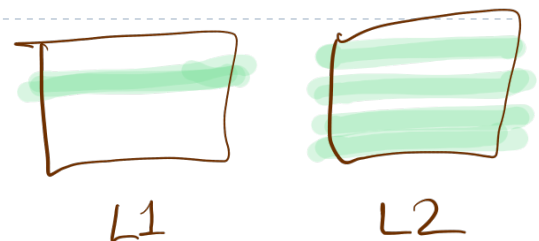
2. Retrieve the CSE students whose CGPA/~~Project marks~~ is higher than all CS students (using ALL).

SELECT * FROM lab WHOSE major = 'CSE' WHOSE cgpa >
ALL(SELECT cgpa FROM lab WHERE major = 'CS');



3. Select those majors for which (at least 1 student has CGPA lower) than 3.7 (using EXISTS).

SELECT DISTINCT major FROM lab L1 WHERE EXISTS(
SELECT * FROM lab L2 WHERE L2.major = L1.major
and L2.cgpa < 3.7);



4. Select name and major from lab where project marks = (select Max(marks) from lab) and cgpa = (select Min(cgpa) FROM lab);

Retrieve name and major of students who have the maximum project marks and minimum cgpa